

Cloud Computing Technologies

DAT515 - Fall 2025

Project Introduction

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Project Overview

- Purpose: Practical application of cloud computing concepts
- Freedom: Choose any cloud technology/service based on your interests
- Duration: Until mid-November
- Team-based: Collaborative development with individual accountability

1. Design Document (design.md)

- Brief, well-organized system design
- Architecture, technologies, deployment strategy
- Updated throughout development

2. Codebase

- Complete source code and configuration files
- Well-organized and documented
- Fully reproducible

3. Reproducibility Report (report.md)

- Step-by-step deployment instructions
- Prerequisites, build, testing, usage examples
- Individual accountability: Hour tracking and self-assessment
- Link contributions to specific GitHub commits

4. Presentation Video (8-10 minutes)

- YouTube upload (public/unlisted)
- Project demo, architecture walkthrough, deployment showcase

5. Q&A Session

- Presentation delivered as a YouTube video
- Criteria review against the checklist
- Further details later

Project Selection

Project Categories

- **Infrastructure & DevOps**

- QuickFeed Continuous Deployment
- QuickFeed Kubernetes Testing
- HotStuff CI Framework

- **Business Applications**

- Stock Exchange Platform
- Personal Finance Tracker
- E-commerce Platform
- Time Tracking & Payroll

Project Categories

- **Communication & Social**

- Team Collaboration Tool
- Social Media Platform
- Local Student Marketplace

- **Media & Gaming**

- Video Streaming Application
- Gaming Platform
- File Storage Service

Your own idea!

Key Requirements

- **Technical**

- Cloud deployment mandatory
- Modern web technologies
- Scalable architecture
- Proper authentication

- **Documentation**

- Reproducibility is critical
- Anyone should be able to deploy your app
- Include troubleshooting guides
- Template provided for report structure

- **Individual Accountability**

- Hour tracking for each team member
- Self-assessment tables
- GitHub commit attribution
- Fair contribution verification

- **Technical Excellence**

- Code quality and organization
- Deployment simplicity
- Architecture decisions
- Cloud technology utilization

Getting Started

- **Form teams** and choose project idea
- Write design document (design.md)
- Get assigned teaching staff contact
- Start development with regular check-ins

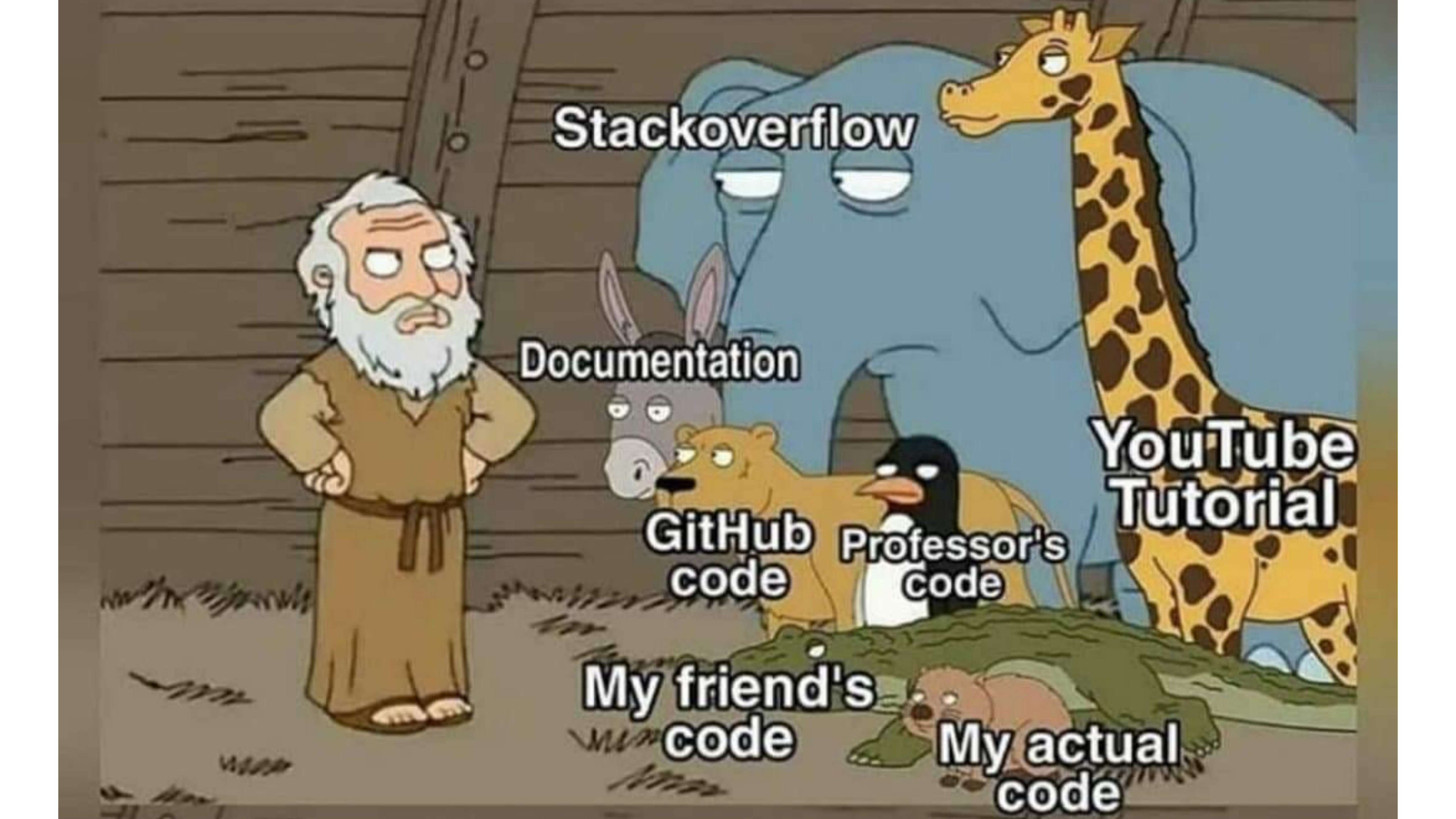
AI Tools Encouraged

- Add AGENTS.md file to guide AI assistance
- Use for code generation, documentation, testing
- Team responsible for final quality and accuracy

Resources

- Course website:
 - Reading Material: Blogs and Documentation
 - Research Papers
- Project evaluation checklist available
- Provided: Templates for design doc and report
- Schedule: Regular meetings with teaching staff

Meme time

A cartoon illustration of Noah's Ark. Noah, an elderly man with a white beard and a brown robe, stands on the left with his hands on his hips, looking towards the animals. The ark is a wooden structure with horizontal planks. Various animals are inside: a large blue elephant with its trunk curled, a tall giraffe, a grey donkey, a brown bear, a penguin, and a small brown dog. Text labels are placed near the animals: 'Stackoverflow' above the elephant, 'Documentation' above the donkey, 'YouTube Tutorial' next to the giraffe, 'GitHub code' above the bear, 'Professor's code' above the penguin, 'My friend's code' above the dog, and 'My actual code' below the dog.

Stackoverflow

Documentation

**YouTube
Tutorial**

**GitHub
code**

**Professor's
code**

**My friend's
code**

**My actual
code**

Meling

my working
code

What the hell is this?





AHA
Session!